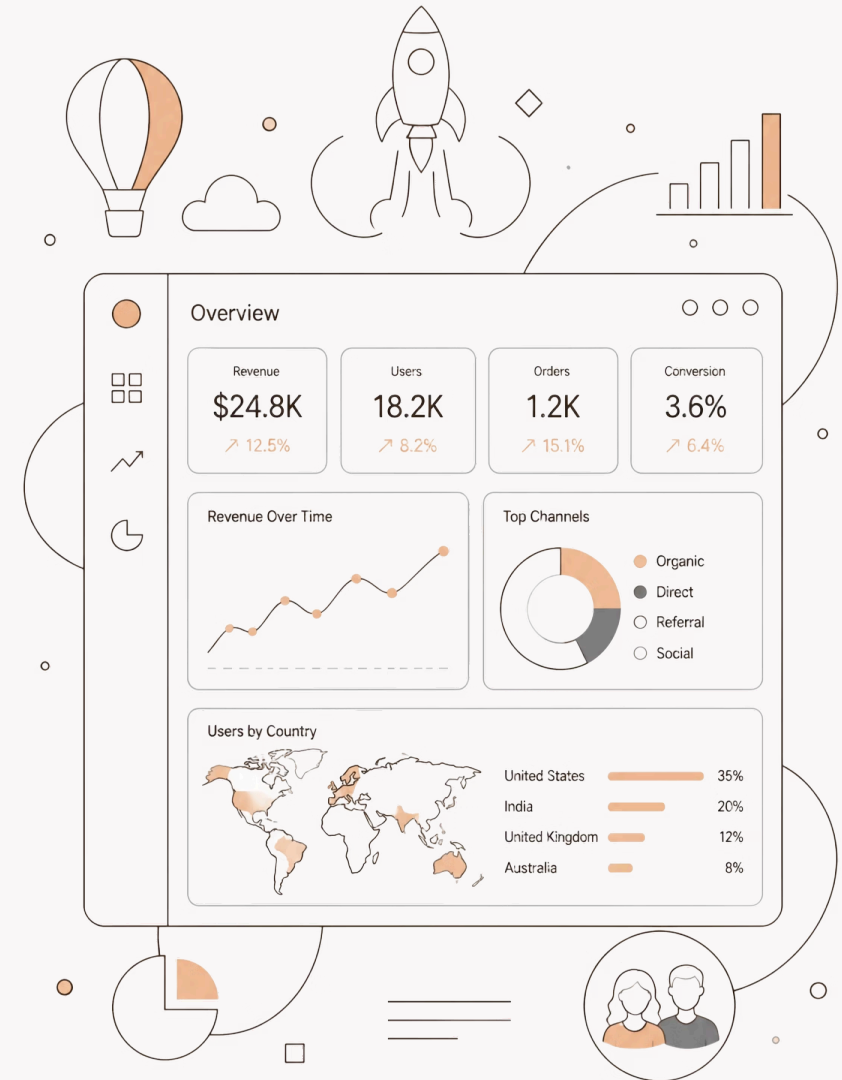


Ride Booking Analysis

End-to-end data analytics project (Version 1.0) — Prepared by ABHISHEK ANTHONY — Tools: Excel, Python, SQL, Power BI, AI Tool — July 2026





Executive Summary

Goal

Analyze booking behavior, cancellations, vehicle performance, and surface actionable insights for decision-making.

Workflow

Data collection → cleaning → EDA (Python) → SQL analysis → Power BI dashboards.

Outcome

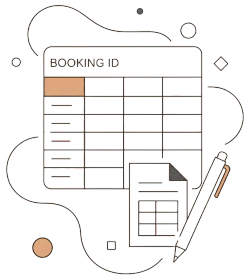
Interactive dashboard + KPIs to monitor performance and guide operational improvements.

Business Problem

Raw booking data hides patterns: cancellations, peak demand, vehicle preferences, payment behavior. The challenge: convert raw logs into insights that reduce cancellations, optimize fleet, and improve customer satisfaction.

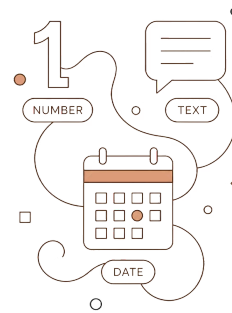


Dataset Overview



Core Fields

Booking ID, Status, Customer ID, Vehicle Type, Pickup/Drop, Booking Value, Distance, Payment Method, Rating, Cancellation Reason, Date & Time.



Data Types

Numerical, Categorical, Date/Time, Text
— suitable for EDA, SQL aggregation, and visualization.

Data Cleaning & Preparation

01

Inspection

`df.head()`, `df.info()`, `df.describe()` to understand structure and issues.

02

Missing & Duplicates

Handled nulls appropriately and removed duplicates to ensure accuracy.

03

Types & Outliers

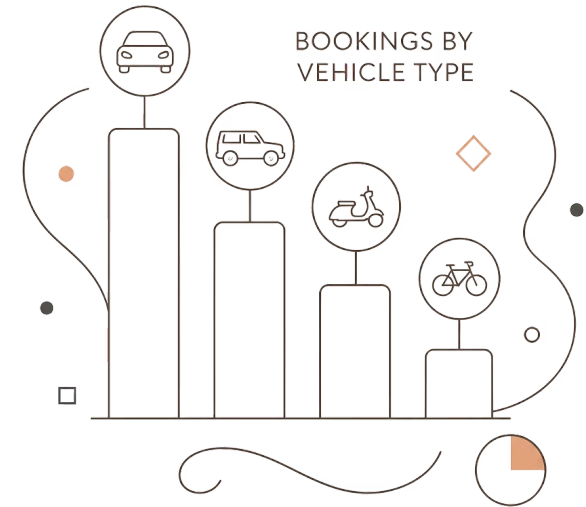
Converted dates/numerics, standardized categories, treated outliers (IQR) to preserve integrity.



Exploratory Data Analysis (EDA)

EDA objectives: distribution of bookings, vehicle preferences, booking values, distances, payment methods, ratings, cancellations, peak times, and high-demand locations.

- Tools: Pandas, NumPy, Matplotlib
- Key finding: demand varies by hour and location; certain vehicle types dominate bookings.



SQL Analysis Highlights

1

Total Bookings

Aggregate booking volume used to measure platform activity and trends.

2

Booking Status & Cancellations

Distribution analysis to calculate cancellation rate and operational impact.

3

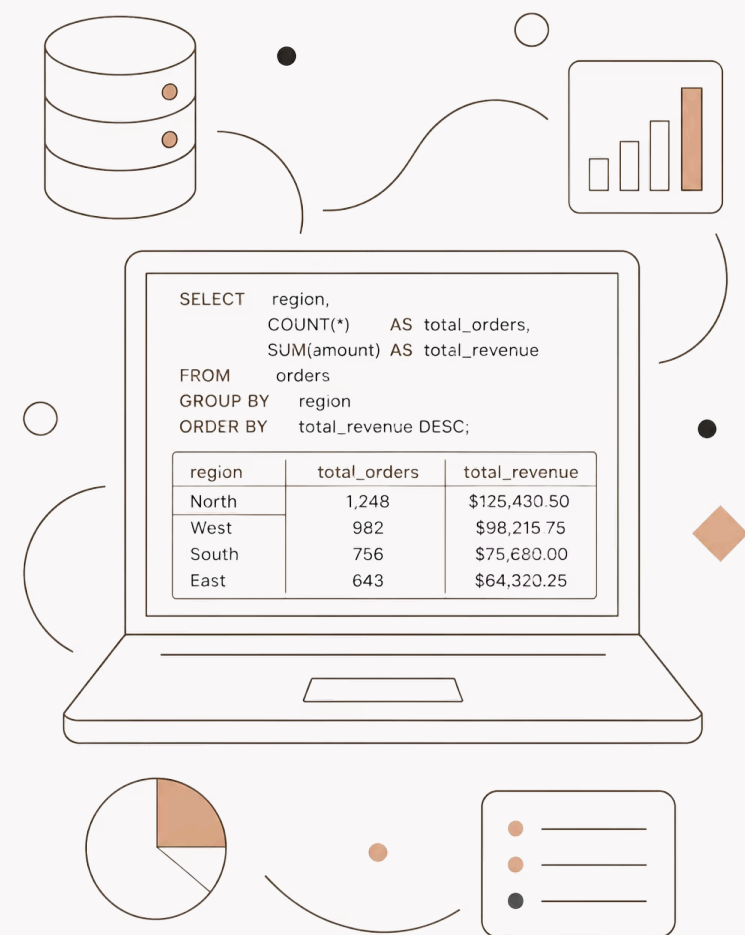
Location & Payment Insights

Top pickup/drop locations and most-used payment methods for operational planning.

4

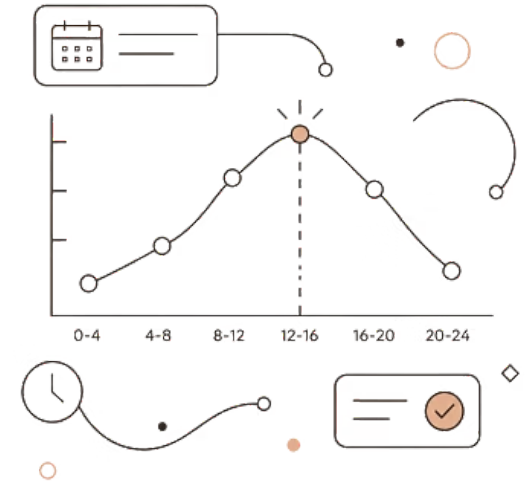
Peak Hours & Ratings

Hourly demand patterns and rating variation across vehicle types to guide staffing and training.



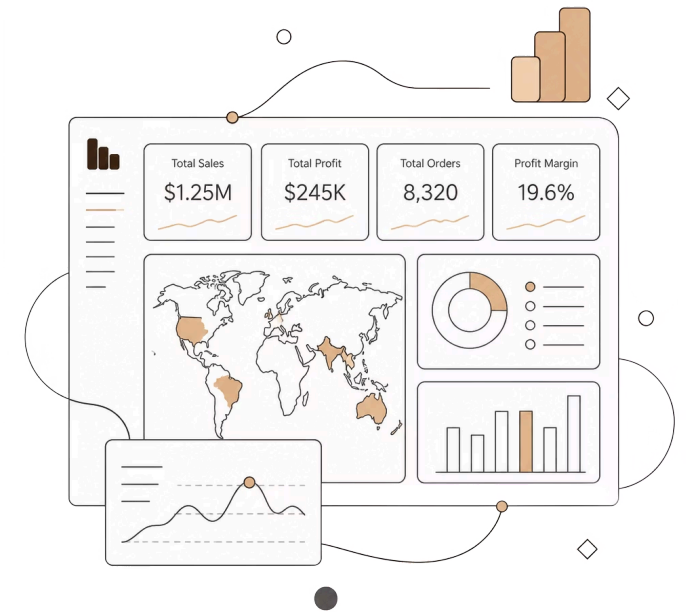
Python Analysis & Visualizations

Python automated cleaning, EDA, and charts (bar, pie, hist, line). GroupBy, aggregation, distribution plots, and visual checks ensured insights were reproducible and exportable to Power BI.



Power BI Dashboard & Business Insights

The dashboard presents KPIs (Total Bookings, Revenue, Completed Rides, Cancellation Rate, Avg Rating) with interactive slicers, drill-downs, and cross-filtering. Key insights: high-demand locations, dominant vehicle types, peak hours, uneven payment preferences, and cancellation drivers.



Recommendations & Next Steps

1

Driver Availability

Increase active drivers during peak hours and in high-demand locations.

2

Reduce Cancellations

Root-cause analysis, improved matching, and clearer customer communication.

3

Optimize Fleet

Allocate vehicles based on category demand to improve utilization.

4

Promote Digital Payments

Incentivize preferred payment methods to streamline transactions.

5

Future Enhancements

Build predictive models, real-time dashboards, customer segmentation, and integrate weather/traffic data.